

Utrecht University

Thesis: Facial attractiveness and Gaze

Title: Who do we look for? The interplay of Attractiveness, Gaze Direction and Observer Identity in Visual Attention

Name: Elena Kajmak Gazideda Rodiqi

Student number: 0069116

Supervisor: David Terburg

Please note: This abstract along with the thesis title are not the final version, they still require finishing touches.

Abstract

Facial attractiveness plays a crucial role in guiding attention, yet its influence on gaze-directed visual search remains unclear. Building on evidence that faces with direct gaze repeatedly orient direction, the present study investigates whether gaze is modulated by facial attractiveness. Participants (N=34) completed a visual search task requiring identification of a target face with direct gaze among distractors with averted gaze. Using standardized stimuli from the Oslo Face Database, with varying attractiveness levels and sex. Eye-tracking data were used to assess whether attention is preferentially allocated to attractive faces, regardless that the task forces attention on gaze direction. Participants' gender significantly influenced visual attention patterns, with consistent effects observed across reaction time, dwell time, and time to first target fixation. A secondary aim examined whether attraction toward the target's gender influenced search performance. Sexual preference was found to have a more limited effect, with significance only for dwell time. Suggesting gender related attentional biases during initial target evaluation, while the influence of sexual preference appears more conservative.

Keywords: Attractiveness, Attention, Eye-Tracking, Gaze-Cueing, Visual Search

Name: Betty (Yuexi) Zhu

Thesis

Title Project: The Influence of Iconographic and Formalistic Instruction on Art Viewing

Behavior

Abstract

This study examines how different art analysis instructions influence visual attention and interpretation during art viewing. Two dominant methods were compared: iconographic analysis, which focuses on symbolic and narrative meaning, and formal analysis, which emphasizes perceptual features such as light, texture, and spatial composition. Thirty participants were randomly assigned to one of the two instructional conditions and viewed ten paintings while their eye movements and verbal descriptions were recorded. Areas of Interest (AOIs) were defined for each painting to distinguish iconographic and formalistic elements. Results showed that participants fixated significantly longer on AOIs congruent with their assigned instruction. The formal group also exhibited more saccades, suggesting a more exploratory viewing pattern. Verbal descriptions aligned with the instructional framing, reflecting either symbolic or perceptual emphasis. These findings provide empirical evidence that brief educational framing can shape both gaze behavior and interpretive strategies in art perception.

Name company or University: Utrecht University

External supervisor: Maarten Wijntjes, Zsafia Pilz

UU Supervisor: Maarten van der Smagt

Name: Betty (Yuexi) Zhu

Thesis

Title Project: The Influence of Iconographic and Formalistic Instruction on Art Viewing

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Name company or University: Utrecht University

External supervisor: Maarten Wijntjes, Zsafia Pilz

UU Supervisor: Maarten van der Smagt

Name Student: Lars van der Haar

Internship or Thesis: Thesis

Title project: To what extent do top-down and bottom-up approaches influence sustainable commuting behaviour in urban professionals?

(Abstract, 100-150 words)

Commuting behaviour among urban professionals remains a significant contributor to transport-related emissions despite their potential for change. This study investigates how top-down (policy, infrastructure) and bottom-up (personal motivation, social influence) strategies affect sustainable commuting behaviour. A quantitative survey (N = 123) with a few open questions measured factors influencing commuting decisions, grounded in the Behavioural Transition to Sustainability (BTS) framework. Structural factors, especially (multiple) car ownership and urban living environment were the strongest predictors of sustainable commuting behaviour. Bottom-up predictors had a weaker influence while it was often high among this group. Effective change requires aligning individual motivation with supportive systems.

Name company or University: Goudappel

(If applicable) External supervisor: Iris Aviezer

UU Supervisor: Christoph Strauch

Name Student: John Oswald

Internship

Title project: Enhancing Knowledge Delivery at Velocity Global

This internship will focus on analyzing and improving the customer user experience (UX) as it relates to accessing critical information at Velocity Global. The study will evaluate the differences in knowledge coverage, presentation, and usability between the two platforms to understand what is missing, and will aim to create a new, customer-centered knowledge product experience.

Velocity Global wants to remove MyGuides and replace it with a superior experience so that customers can more efficiently interact with and use critical business information. The project will assess current gaps, explore customer needs, and design an improved knowledge delivery experience aligned with UX best practices.

The primary goal of this internship is to phase out MyGuides and replace it with a refined knowledge experience backed by empirical research. The expected deliverables include a comparative analysis report, user personas, a customer journey map, survey findings, and UX design recommendations.

Name company or University: Velocity Global

(If applicable) External supervisor: Danielle Tappitake

UU Supervisor: Sjoerd Stuit

Name Student: Malte Frerichs

Internship or Thesis: Internship

Title project: Creating a Model to Predict Visual Hemifield and Eye Advantage

(Abstract, 100-150 words)

Binocular rivalry results from competition between both eyes, where perception can alter; it is often also referred to as interocular competition. During binocular rivalry, one eye's image dominates visual perception, leading the other to be suppressed. One facet of interocular competition that has received only brief attention in other papers is the visual hemifield in which the visual stimuli are presented. The goal of this research is to correctly predict which eye and visual hemifield will have the advantage in different situations. It is hypothesized that the eye and visual hemifield that can view the majority of information will hold the advantage. To achieve this, a program was created that can calculate the viewable percentage of a target image for each eye and visual hemifield. This program rests on the assumption that the eye or visual hemifield with more information has the advantage. Moving forward, this program will be tested experimentally from which on improvements to the program as well as the underlying theory can be made.

Name of Company or University: University

(If applicable) External supervisor:

UU Supervisor: Chris Paffen

Name Student: İpek Atlıhan

Internship or Thesis: Thesis

Title project: Designing an Adaptive Learning Interface Using Real-Time Webcam-Based Eye-Tracking and Cognitive Load Theory

Abstract

This thesis investigates the development and evaluation of a webcam-based adaptive learning interface designed to respond to learners' real-time cognitive states. Grounded in Cognitive Load Theory (CLT), the system uses eye-tracking metrics such as fixation duration, blink rate, regressions, and gaze direction to infer cognitive load and adjust the interface accordingly. Implemented within the CoVince platform, the adaptive interface dynamically modifies layout, modality (e.g., audio vs. visual), and scaffolding. A within-subject design compares performance, cognitive load, and satisfaction between adaptive and static conditions. Subjective workload is assessed using NASA-TLX, and satisfaction is evaluated with a 5-point Likert scale. Results are expected to provide insights into the effectiveness of gaze-driven adaptivity in enhancing engagement and learning outcomes in online educational environments.

Name company or University: CoVince

External supervisor: Frederike Zwenk

UU Supervisor: Fleur van Ierschot

Name Student: Robert Kwakernaat

Internship or Thesis: Thesis

Title project: Too Powerful to Stop? The Role of Theta Oscillations in Response Inhibition

Abstract:

The ability to inhibit a response is crucial for behavioral flexibility. Frontal theta power, the strength of the rhythmic neural oscillations within the range of 4-8 Hz, is associated with successful proactive response inhibition, which prepares sensory-motor connections to inhibit. The objective of the current study is to replicate this finding.

To measure response inhibition, the stop signal task (SST) was used. In the go trials, participants had to press a key when a stimulus was shown. In the stop trials, participants had to inhibit their response to the stimulus when a sound was played. During the task, EEG data was recorded.

No significant difference in theta power in the Fz-electrode was found between successful and failed inhibitions, which contrasts earlier findings. The results suggest that the effect may not generalize reliably across different samples. Further research is needed to clarify the exact function and occurrence of theta oscillations in response inhibition.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Sam van Bijnen

Name: Zsófia Varga

Internship or Thesis: Internship

Title: Design Principles for HMI Used Under Stress

This internship project, conducted at Thales Nederland B.V., contributed to the EU founded CHIMERA (Comprehensive Hazard Identification, and Monitoring systEm for uRban Areas) application development by exploring the effects of stress on human cognition and how HMI (human-machine interface) design strategies can mitigate these effects. The CHIMERA system aims to support first responders during CBRN (chemical, biological, radiological and nuclear) incidents, by improving technological capabilities and situational awareness of the end-users. A literature review was conducted to identify the main cognitive effects of stress, focusing on attention, memory, strategic thinking, and decision making. Based on the findings, mitigation strategies were compiled, that were later applied to the application interface design.

Name of company or University: Thales Nederland B.V.

External Supervisor: dr. Iris Cohen

UU Supervisor: Prof. dr. Leon Kenemans

Student: Nele-Rita Põldma

Internship/Thesis: Thesis

Project Title: **Investigating the Structural Correlates of Empathy: A VBM and MVPA approach**

Empathy is a multidimensional construct essential to social functioning, yet its structural neural correlates are not well understood. This thesis investigated how individual differences in self-reported empathy relate to brain structure in a sample of 242 healthy adults, using voxel-based morphometry (VBM) and multivariate pattern analysis (MVPA). Empathy was measured with the Interpersonal Reactivity Index (IRI), which includes four subscales: Perspective Taking (PT), Empathic Concern (EC), Fantasy (F), and Personal Distress (PD). VBM revealed that higher PT scores were associated with increased grey matter volume in the left cerebellum and decreased volume in the right temporal pole. No significant associations were found for EC, F, or PD. MVPA failed to classify empathy levels above chance, suggesting that local grey matter patterns may not reliably capture individual differences in empathy. These findings highlight the involvement of cerebellar and temporal regions in cognitive aspects of empathy and point to the complexity of its structural representation.

Company/University: Utrecht University

UU Supervisor: Joana B. Vieira, PhD

Name Student: Navya Suneej

Internship or Thesis: Internship

Title project: What role do anticipated outcomes play in behaviour, thereby distinguishing between habits and goal directed behavior?

The research project will be regarding habitual versus goal directed control of behaviour. Specifically, we want to know what happens when people are trained on an instrumental learning task to make specific responses to obtain a reward. According to a dual process model of decision, by repeatedly conducting this behavior a stimulus response habit will form and the anticipation of the outcome becomes increasingly less relevant. Previous studies that have used outcome-devaluation to test whether behavior becomes less driven by its anticipation of the outcome does have some limitations. Therefore, we would like to use a different paradigm using bi-stable perception to obtain an implicit measure of the strength of the outcome anticipation similar to bi-stable spheres.

Name company or University: Utrecht University

(If applicable) External supervisor: Tim van Timmeren

UU Supervisor: Jana Klaus

Name Student: Fan Yang

Internship or Thesis: Internship

Title: To Alarm, or Not to Alarm? Assessing Alarm Fatigue in Nurses

Intensive Care Units (ICUs) are critical environments where patients require constant monitoring. Rapid detection of alarms is essential for saving patients' lives. However, a high number of clinically irrelevant false alarms can lead to desensitization, causing caregivers to ignore alarms. Current study aims to explore relationships between alarm fatigue, perceived stress, noise sensitivity, soundscape assessment, personality profile and perceived workload among ICU nurses, with the goal of reducing alarm fatigue on nurses. A total of 16 surveys were collected from the nurses. No statistically significant relationships were found between demographics and measured variables. The average alarm fatigue score was 57.9 ± 25.3 , indicating a moderate level of fatigue. Participants reported high noise sensitivity (18.19 ± 2.10) and described the acoustic environment as being dominated by medical device sounds (3.53 ± 0.83). Overall, alarm fatigue levels remained moderate, while ICU nurses demonstrated high sensitivity to noise. Additional data will be collected to support further analysis.

Name company or University: Critical Alarms Lab (TU Delft & Erasmus MC)

External supervisor: Elif Ozcan & Margo van Mol (Erasmus MC)

UU Supervisor: Anouk Keizer

Abstract Update Day 3, Internship Thes Groeneveld

Name Student: Thes Groeneveld

Internship or Thesis: Internship

Title project: Designing a Semi-Structured Interview to Explore Stakeholder Perspectives on Nudging-Based Features for Safe Walking Paths for Elderly Residents with Dementia

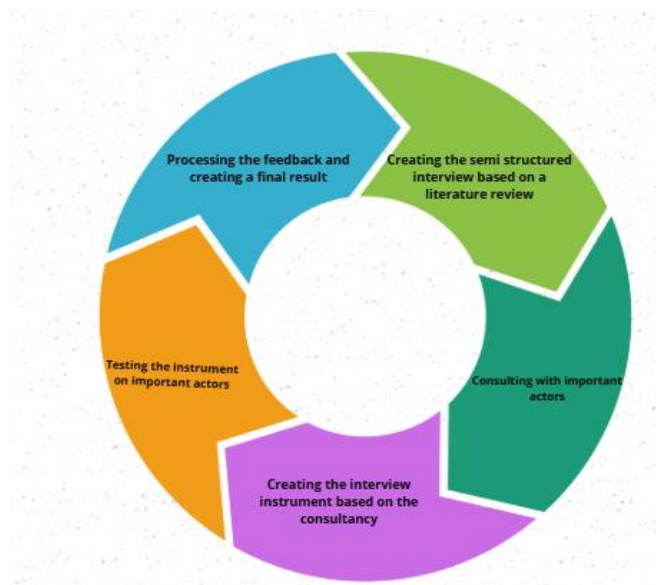
(Abstract, 100-150 words)

Name company or University: Stichting QuaRijn

(If applicable) External supervisor: Dr. Sjenny Winters

UU Supervisor: Dr. Albert Postma

This internship project aims to contribute to the research on designing walking paths that help navigate residents with dementia safely in the outdoor environment, utilizing nudging techniques that can be implemented within this environment. The project adds to this research by creating the first step of designing a questionnaire, an interview instrument, for four different stakeholders (= ergophysiotherapists, informal caregivers, formal caregivers, and design experts). First, a literature review identifies key elements that can be questioned. For the second step, stakeholders are asked to think of questions. The verbatims are used to make a concept for the interview instrument. Then there will be four new interviews with new stakeholders within the same groups mentioned before to review the instrument. The interview instrument will be perfected. Future research can continue research based on the created instrument.



Name Student: **Sarah Vollmer**

Internship or Thesis: **Thesis**

Title project: **Psychophysiological responses to active and passive threat avoidance – a balance board study**

Threat responses are understood to operate on the threat imminence continuum (Fanselow, 1994). Distal threats are associated with attentive freezing (reduction of body sway, bradycardia, increase in the startle response), while imminent threats are characterised by a decrease of the startle reflex and an increase in heart rate and skin conductance response (Mobbs, 2009). It is unclear how having the ability to avoid threats influences this continuum, as studies have argued that freezing is more pronounced when the threat is inevitable (Löw et al., 2015), while others have found the opposite (Merscher & Gamer, 2024).

The current study utilized a threat avoidance task, in which participants could either avoid a shock or had to passively endure it. Skin conductance, heart rate, electromyographic, startle responses, and body sway were measured. It is hypothesized that attentive freezing will be augmented in the avoidable threat condition, alongside a potentiation of the startle response.

Name company or University: **Utrecht University**

UU Supervisor: **Joke Baas**

Fanselow, M. S. (1994). Neural organization of the defensive behavior system responsible for fear. *Psychonomic bulletin & review*, 1(4), 429-438. <https://doi.org/10.3758/BF03210947>

Löw, A., Weymar, M., & Hamm, A. O. (2015). When threat is near, get out of here: Dynamics of defensive behavior during freezing and active avoidance. *Psychological science*, 26(11), 1706-1716. <https://doi.org/10.1177/0956797615597332>

Mobbs, D., Marchant, J. L., Hassabis, D., Seymour, B., Tan, G., Gray, M., Petrovic, P., Dolan, R. J., & Frith, C. D. (2009). From Threat to fear: The neural organization of defensive fear systems in humans. *Journal of Neuroscience*, 29(39), 12236–12243. <https://doi.org/10.1523/jneurosci.2378-09.2009>

Merscher, A. S., & Gamer, M. (2024). Can I see it in the eyes? An investigation of freezing-like motion patterns in response to avoidable threat. *Psychophysiology*, 61(7), e14567. <https://doi.org/10.1111/psyp.14567>

Name Student: Alina Aravand

Internship or Thesis: Thesis

Title project: Trait Impulsivity as a Pathway Linking Cannabis Use Severity to Response Inhibition Deficits.

This study explored the interactions between cannabis use severity, trait impulsivity, and cognitive control, with a particular focus on response inhibition. Participants from a general population sample completed the Cannabis Use Disorders Identification Test (CUDIT) and Barratt Impulsiveness Scale (BIS-11) and performed a stop-signal task while EEG was recorded. In a subsample of 18 cannabis users ($N = 46$), exploratory mediation analysis revealed a significant indirect effect of cannabis use severity on stop-signal reaction time (SSRT) via trait impulsivity, with no significant direct effect. These results suggest that greater cannabis use may impair response inhibition by increasing impulsivity. Although impulsivity is often viewed as a stable trait, findings suggest it may be influenced by behavioural factors such as substance use. The results highlight a potential pathway linking cannabis use to cognitive control deficits, informing prevention and intervention strategies. Ongoing EEG analyses examine P3 event-related potential differences between smokers and non-smokers to better understand neural mechanisms of response inhibition.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Daan van Rooij

Name Student: Jippe Jansman
Internship or Thesis: Thesis
Title project: fNIRS and the future of mental fatigue monitoring in aviation

Mental fatigue (MF) is a prevalent problem in aviation, especially when considering the increasing automation. It occurs, for example, during long periods of low mental effort, which often happens in the cockpit. This study investigates functional near-infrared spectroscopy (fNIRS) as a reliable indicator of MF using subjective rating scales as a ground truth. Participants took part in a 70-minute simulated flight with stages with varying difficulty. We measured their fatigue levels using subjective rating scales and brain oxygenation, and we measured performance. Participants reported feeling more fatigued during the prolonged easy stage, this rating declined with increasing task difficulty. The fNIRS data showed conservative changes in HbO levels, which were somewhat connected to subjective fatigue ratings. However, we failed to see a clear link between HbO levels and performance. While fNIRS could be useful for monitoring MF, it is unclear if HbO concentration can predict performance in this setting. Future studies should consider more demanding tasks, enhanced signal processing, and larger samples to better understand the practical applications of fNIRS in aviation.

Name company or University: Royal NLR
(If applicable) External supervisor: Carmen van Klaren
UU Supervisor: Gabriël Beckers

Name Student: Elif Nur Keskinilic (2107376)

Internship or Thesis: Internship (12.5 ECTS)

Title project: Redesigning Buddy: Phased, Guided Feedback to Improve Mental Health Report Usability

This internship project focuses on redesigning the AI-powered “Buddy” chatbot within CoVince’s NEXUS platform, which delivers personal development reports based on the VRR (Vulnerability, Resilience, Risk) model. Although these reports provide detailed insights into psychological factors such as anxiety, resilience, and stigma, their dense structure can overwhelm users, especially those without a background in mental health. Guided by Cognitive Load Theory, the project introduces the “Buddy Guide”, a phased feedback structure that presents results step-by-step using progressive disclosure. This approach aims to reduce cognitive overload, enhance user understanding, and create a more emotionally supportive experience. The redesigned chatbot structure includes intro messages, graph-specific explanations, and follow-up questions tailored to the user’s flow. The current phase of the project involves finalizing this structure. Once completed, user testing will be conducted to assess whether the Buddy Guide improves comprehension, usability, and perceived support compared to the original format.

Name company or University: CoVince

(If applicable) External supervisor: Richard van Tilborg

UU Supervisor: Dr. Samson Chota

Abstract

Student: Tomas Santiago Bonanno (2418312)

Project: Thesis

Title Project: How Brands Can Stimulate Responsible Consumer Behaviour

University: Utrecht University & Delft University of Technology

UU Supervisor: David Terburg

External Supervisor: Sijia Wu Bakker

This thesis explores how brands can serve as a strategic tool to promote responsible consumer behavior. The project focuses on mineral sunscreens, which—unlike most conventional chemical sunscreens—do not harm marine ecosystems and may be safer for human health. Despite these benefits, mineral sunscreens face aesthetic barriers that limit widespread adoption. To address this, the project implements a communication strategy grounded in cognitive psychology, integrating storytelling and priming techniques that target individual self-perception. By activating identity-relevant values, such as empathy, responsibility, and sustainability, the brand narrative reframes sunscreen use as an act of self-expression and environmental agency. This approach seeks to generate symbolic brand value capable of influencing perception and motivating behavioral change. The central hypothesis proposes that brand messages appealing to the individual self are more effective in promoting the adoption of mineral sunscreens than negatively emotional or rational appeals. To test this, a within-subjects experiment ($N = 60$) was conducted in which participants were exposed to six randomized brand messages (2 self-based, 2 negatively emotional, 2 rational). Following each message, participants rated purchase intention, identity alignment, and perceived agency. Results indicate that self-based messages consistently outperformed the others across all variables, suggesting that identity-primed storytelling can effectively support sustainable decision-making and facilitate pro-environmental behavior change.

Name Student: Anastasia Caraus

Internship or Thesis: Thesis

Title project: Global User Experience Team: Enhancing Effectiveness

This study explores how virtual User Experience (UX) teams can enhance collaboration processes to foster effective team behavior and improve outcomes. Conducted within Capgemini's UX Tower, a cross-functional, globally distributed team of UX professionals, the research addresses the growing need for context-specific insights into virtual team functioning. While virtual teams offer benefits such as flexibility and access to diverse talent, they also face significant challenges, including limited interpersonal connection, trust issues, and barriers to effective information sharing. These obstacles are especially critical in UX work, which relies on creativity, feedback, and shared understanding.

To investigate these dynamics, the study employs a mixed-methods approach. The quantitative phase utilizes a survey based on Dekker's (2008) Effective Virtual Team Behavior (EVTB) framework to assess collaboration practices and also measures trust, satisfaction, and performance. Complementing this, qualitative data are collected through semi-structured interviews using the Critical Incident Technique (CIT) to uncover nuanced interaction behaviors and experiences. Findings reveal key enablers—such as idea-sharing during brainstorming—and barriers, including inaccessible information channels and reduced trust due to the lack of face-to-face interaction. The study offers practical recommendations for improving virtual collaboration and contributes to the broader understanding of effective virtual team behavior in UX contexts.

Name company or University: Capgemini

(If applicable) External supervisor: Christina Veelenturf

UU Supervisor: Sjoerd Stuit

Name Student: Julia Donker

Internship or Thesis: Master Thesis

Name company or University: Netherlands Aerospace Centre (NLR)

External supervisor: Carmen van Klaren

UU Supervisor: Chris Paffen

Title project: The impact of time Pressure on stress responses and decision-making performance in military aviation

Abstract

Decision-making in military aviation often occurs under intense time pressure, with potentially life-threatening consequences. This study investigated how time pressure affects physiological and subjective stress responses, and whether these in turn influence performance outcomes such as reaction time, confidence, and decision accuracy. Twenty-five participants completed a simulated Quick Reaction Alert (QRA) mission in a Generic Fighter Simulator (GFS), under time pressure and no-pressure conditions. Results showed that time pressure significantly increased heart rate and reduced heart rate variability, indicating elevated physiological stress. Participants also reported higher subjective stress and workload. Despite faster reaction times under pressure, decision confidence significantly decreased, while accuracy remained unaffected. These findings underscore the complex interplay between stress and performance, and highlight the need for stress-informed training and interface design in high-stakes aviation contexts.

Student: Daniel Benavides Oviedo— 2769689

Thesis

Title: Is AI really the better teacher? The effect of rule-based and open-speech embodied conversational agents on learning and performance in dialogue skills.

Abstract:

Training adults in conversational skills requires prior knowledge and repeated practice. Embodied conversational agents offer safe, trial-and-error simulations that resemble real-life interactions. Two design models are commonly used: (1) rule-based, involving predefined multiple-choice interactions, and (2) generative (AI-powered), where learners speak flexibly without predefined paths. Compared to generative models, rule-based systems may better support learning and self-efficacy in learners with limited prior knowledge. However, they may be less engaging, potentially limiting learning.

This study explores whether there are differences in learning and performance between the two models, and whether prior knowledge moderates their effects on learning and self-efficacy.

Participants (n = 49 so far) were randomly assigned to either the rule-based or generative condition and completed five simulations. Runs 1–4 involved repeated practice with condition-specific avatars; Run 5 was a shared generative transfer task. Self-efficacy and perceived task complexity were measured before, during, and after the simulations.

Analyses will include two-tailed t-tests comparing learning slopes and transfer performance, and ANOVAs examining prior knowledge (initial performance for run 1) as a moderator. Results are currently being analyzed.

Utrecht University

UU Supervisor: Dr. Maarten, van der Smagt

Update Day Poster Information

Name Student: Feline van Gelderen

Internship or Thesis: Internship

Title project: Blue or Gold? Gaze Allocation and Perception of the Lindau Crucifixion

The Lindau Crucifixion is an early fifteenth-century panel painting that shows Christ on the Cross with Mary and St John the evangelist mourning his death, while four small angels catch his blood in golden chalices.

It was recently discovered that the darkblue background layer was painted on approximately a century after the conception of the painting, and obscures an originally golden background.

AIM: To explore the differences between the two versions of the painting, using digital reconstructions.

Experiments:

1. Gaze allocation (where do people look?): eye tracking (in the lab)
2. Perception (what do people see?): textual description (at home)

The two versions of the painting are presented to all participants among other similar medieval Christian paintings, counterbalanced in order.

RELEVANCE: May attest to feasibility and usefulness of applying these 'modern' techniques in future restoration dilemmas.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

'External' (also UU) Supervisor: Sanne Frequin

Name Student: Havva Irem Aslan

Internship or Thesis: Internship (12,5 ECTS)

Title Project: Recognizing Music-Induced Emotion Mechanisms: A systematic review of cognitive and expressive indicators

Abstract

Music, unlike most other stimuli, is intentionally designed to induce emotions. Yet, designing experiments to evoke specific emotions is complex due to the various psychological mechanisms that influence the relationship between listeners and music. Specifically, researchers have often studied music-induced emotions without considering how these emotions were elicited. This systematic review examines emotional responses to music, focusing on subjective feelings and bodily expressions, such as facial reactions. It also examines induction mechanisms, such as emotional contagion, that can arise from different stimuli. Following PRISMA guidelines, literature searches were conducted in major databases such as ScienceDirect and screened using the active learning tools ASReview and Litmaps. Methodological diversity in facial analysis tools and emotion models further contributes to variations in reported correspondence. By synthesizing theory and empirical findings from multiple domains, this review highlights the importance of integrating multimodal emotion measures to design experiments that position music as a proactive stimulus, enabling researchers to capture aligned bodily and cognitive responses. Such insights can support applications in music therapy, user experience design, and marketing.

Name company or University: Utrecht University

UU Supervisor: Marnix Naber

name: Richa Zope (4888286)

project: Thesis

title: Exploring the behavioral effects of sensorimotor integration in Neuralgic Amyotrophy with the Rubber Hand Illusion

Neuralgic amyotrophy (NA) is a peripheral nerve disorder that often leads to long-term motor issues despite peripheral recovery. Recent evidence suggests that altered multisensory integration may underlie these persistent symptoms. This study uses the Rubber Hand Illusion (RHI) to assess proprioceptive drift, a behavioral marker of multisensory integration, in NA patients. We compare drift between affected and unaffected limbs, and against healthy controls, to explore links with upper limb function. In addition, we explore possible interactions between drift and upper extremity function. Data of 13 patients and 10 healthy controls have been collected. Still two healthy participants need to be measured to be able to fully analyze the data. A 2x2x2 mixed design repeated measures ANOVA will be performed to investigate differences in proprioceptive drift between the dominant and non-dominant hand, between synchronous and asynchronous stimulation and between patients and healthy controls. A Pearson correlation will be performed to investigate a possible correlation between proprioceptive drift and upper extremity function.

Utrecht University

UU supervisor: Renee Lustenhouwer

Femke van Nielen- Thesis

The temporal development of motion sickness symptoms and general feelings of sickness

TNO integrated vehicle safety department

External supervisor: Jan Souman

Internal supervisor: Leon Kenemans

Abstract

To better understand and mitigate motion sickness, it is essential to be able to quantify its progression during motion exposure unambiguously. Self-report rating scales either focus on measuring specific symptoms or on general feelings of unpleasantness or sickness. Previous research has suggested that motion sickness symptoms generally manifest in a fixed order, with non-nausea symptoms preceding nausea. In contrast, unpleasantness was shown to increase non-monotonically, with a temporary reduction at the transition from non-nausea to nausea symptoms. However, other authors have suggested that unpleasantness does increase monotonically with symptom progression. This thesis aims to clarify the relationship between feeling of unpleasantness and symptomatology and additionally conclude if non nausea symptoms only precede nausea.

Participants were exposed to the same provocative motion stimulus in two different sessions on separate days. In one session, participants alternately rated their symptoms using the Motion Illness Symptoms Classification scale (MISC) and the experienced sickness using the Fast Motion sickness Scale (FMS) with 30 s intervals in between. In the other session, participants rated the extent of both non-nausea and nausea symptoms they experienced on separate scales, in alteration with FMS ratings.

The results will provide insights into the progression of motion sickness symptoms over time and how they relate to the level of experienced sickness.

Henrik Stockinger

Presentation on Internship (7.5 ECTS) & Thesis (27.5 ECTS)

Researching the Effects of Caregiver Gestures on Children's Gestural Output and Future Vocabulary

The purpose of my internship and thesis has been to investigate the effects that parental gestures have on their children's gestural output and subsequent linguistic abilities. The internship portion consisted of creating a coding scheme to annotate parental gestures in a subset of 111 videos. After obtaining all the data and conducting a brief analysis of the data, the gestural data of the parents was used alongside previously obtained data from the children in the study to answer two key questions. The first was whether the number of points and "showgives" elicited by parents positively correlated with the amount of those same gestures produced by the children. The second question was whether the amount of non-referential gestures, like object manipulation, can predict the future linguistic abilities of a child. Using JASP, a simple correlation and linear regression model were used respectively. No correlations were found in either case. The potential reasons were investigated via literature review.

Utrecht University

UU Supervisor: Klis, A. van der

Name Student: Hasan Goulichli (Güliçli)

Internship or Thesis: Thesis

Title project: The Influence of Virtual Body Ownership on the Extended Self: Presence as a Mediator and Empathy as a Covariate

(Abstract, 100-150 words)

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Dr. Anouk Keizer

Belk's (1988, 2016) expanded self concept suggests that our identity can encompass external objects, even virtual bodies. However, in virtual reality (VR), the role of sensorimotor focused body ownership, presence, and empathy in shaping this extension remains unclear. In this study (N = 63), participants experienced two VR conditions, synchronous versus asynchronous avatar movement, and then completed measures of extended self, body ownership, presence, and empathy. We analyzed the data using bootstrapped paired-samples *t* tests, Pearson correlations, hierarchical regression, and structural equation modeling (SEM). Consistent with predictions, extended self scores were significantly higher following synchronous than asynchronous movement. Body ownership correlated with both presence and extended self, and SEM revealed a large direct effect of body ownership on extended self but no significant indirect effect via presence. Hierarchical regression showed that empathy did not predict extended self nor alter the ownership on extended self link. These findings show that sensorimotor synchronization strongly drives self-expansion in VR environments, independent of individual differences in empathy, and that presence, although related to ownership, does not mediate this process in short-term exposures.

Name Student: Zoi Pantou 5164702

Thesis

Title project: Examining ERP and Cognitive Measures of Response Inhibition in Hypersensitive Individuals

Name company or University: Utrecht University

UU Supervisor: Dr Daan van Rooij

The aim of the study was to examine the link between response inhibition and sensory sensitivity, using both behavioral and neural markers. Stop-Signal Task (SST) and the Multi-Modal Evaluation of Sensory Sensitivity (MESSY) questionnaire were used to examine if inhibitory control performance, measured by Stop-Signal Reaction Time (SSRT) and neural correlates, indicated by P3 event-related potential, differed among individuals with different levels of sensory sensitivity. The results showed no meaningful connection between SSRT and MESSY scores. This suggests that behavioral measures of inhibition might not reflect differences in sensory sensitivity. There was a trend indicating that individuals with high MESSY scores had faster average reaction times, though this result was not statistically significant. Preliminary EEG analyses focused on fronto-central P3 components, with full results pending to assess if neural markers predict sensory sensitivity better than behavioral measures.

Name Student: Tijani Nagra

Internship or Thesis: Internship

Title project: Advisory Report on Enhancing the Stick Project: Recommendations for Future Development

(Abstract, 100-150 words)

Name company or University: Utrecht University

(If applicable) External supervisor: ---

UU Supervisor: Manja Engel

Abstract This advisory report evaluates the functioning of Project Stick and identifies areas for improvement, particularly in the data analysis and the duration of tactile feedback. The primary goal is to refine data analysis methods and design an experiment to assess variations in tactile feedback duration. The report examines adjustments to data processing, including transitioning from one-dimensional to three-dimensional measurements, establishing an accuracy metric in R, and implementing necessary experimental modifications. Statistical analysis of two existing datasets reveals that three-dimensional measurements introduce noise, indicating that X- and Y-coordinates are a more reliable approach for accuracy assessment than measuring effect of tactile feedback. Additionally, a pilot experiment comparing tactile durations suggests an increasing effect over time, supporting further investigation into sensory integration. To enhance precision, future research should examine the alignment between calibration points and participants' body perception, alongside expanding data collection in tactile duration experiments to refine sensory manipulation techniques.



Thesis

Name student: Nina Verbeek

27.5 ECTS Master's Thesis Applied Cognitive Psychology x Educational Sciences

Title project: "To ISC, or not to ISC. That is the question..."

Do synchronized eye movements predict test scores, or do they merely reflect synchronized processing?

Abstract

As education shifts toward digital formats, pre-recorded lectures have become standard. This eliminates real-time student-teacher interaction and the cues instructors rely on to assess engagement and understanding. Intersubject Correlation (ISC), measured through eye tracking, offers a potential solution by measuring how closely a student's gaze aligns with peers. Higher ISC has been linked to better learning outcomes, suggesting it may reflect effective engagement.

However, ISC assumes shared gaze on relevant content. When gaze is drawn to interesting but irrelevant elements – seductive details – ISC may instead reflect shared distraction, where seductive details are known to disrupt learning outcomes. This study examined whether seductive details undermine ISC's predictive value. Using webcam-based eye tracking, students watched either a Control or a Seductive Details version of a knowledge clip on lightning. The Seductive Details group showed significantly higher ISCs. However, learning outcomes did not differ significantly between groups. The non-significant pattern in test results may reflect a ceiling effect due to the sample of motivated university students, unlike the more diverse sample used in previous research.

External supervisor: Ellen Kok (Educational Sciences, UU)

UU Supervisor: Roy Hessels

Name Student:: Melissa Kragten

Internship or Thesis: Thesis and Internship combined

Title project: . Turn-taking using eyes, arms and words: The influence of robot and user turn-taking cues on conversations

Older conversational systems like ELIZA and modern digital assistants like Alexa rely on the simple turn-taking system based upon gaps of silence between turns. But this question-answering and command-and-control systems are very different from open-domain conversational dialogue, where the user has no expectations of the capabilities and restrictions of the system, it is harder to establish common ground for turn-taking. Thus a system influenced by human turn-taking is needed. Current study will compare robot turn-taking behaviour within a conversational setting. During the experiment, the participants will go through one condition of the following: no movement, random head and arm movement and timed head and arm movement. During the conversations, speech will be recorded to measure the conversation quality (interruptions, overlaps, utterance length) and afterwards a conversational dynamics survey will be filled in by the participants.

Name company or University: ING

(If applicable) External supervisor: Marc Jansen

UU Supervisor: Maartje de Graaf

Name: Sanyukta Nair

Project: Thesis

Expressing Saccade Effort in Monetary Terms

Eye movements can provide insight into attentional selection. Recent research has revealed that saccade selection- the decision of where to move the eyes, involves an inherent effort of planning and executing eye movements (saccade costs). Pupillometry data reveals people have a preference for the most ‘affordable’ movements. This research attempts to corroborate the findings that the way we look is an effort-optimizing process by introducing rewards (points convertible to money). It first addressed whether and how saccade costs and rewards together drive saccade selection using a paid target selection task. Here, participants engaged in reward maximization and avoiding zero reward, and costs were reduced but not eliminated when rewards were offered - being tolerated more the higher the reward. The next study drew from these findings to express the ‘costs’ of saccades across directions in points convertible to money. Participants aimed to maximize rewards in a cued reward selection task and their choices shifted the values across directions towards cheaper points (representing relative affordability) or expensive points (representing greater effort). These final payouts across directions correlated strongly with the pupil-size derived costs across directions, confirming the theory that the way we look is a neuroeconomic decision process, and attentional selection involves the consideration of certain effort costs (minimization) and benefits (reward maximisation) to eye movements.

Utrecht University

Supervisor: Christoph Strauch

Co-Supervisor: Damian Koevoet

Name Student:	Kim van Beusichem
Internship or Thesis:	Thesis
Title project:	The effect of tape colour and pattern of application to indicate signage is out-of-use on comprehensibility in roadworks zones.

Abstract

Understanding how road users respond to roadworks factors is essential for ensuring traffic safety in these zones. One of these factors is temporarily out-of-use permanent signage. This study examined the effect of both tape colour and pattern of application used for marking overhead traffic signs out-of-use on comprehensibility. Participants (N = 1007) completed a computer-based task in which they viewed images of traffic signs that were marked out-of-use by tape and indicated what driving action they would take if the situation shown was real. Comprehensibility was assessed using measures of accuracy, reaction times and confidence ratings. The results show significant main effects of colour and pattern on reaction times and of tape pattern on confidence. Interestingly, significant interaction effects were found for the effects of tape colour and pattern on both reaction times and confidence. Accuracy scores did not differ between conditions. These findings suggest that the features of the tape used to indicate signs are out-of-use play an important role in how quickly and confidently drivers comprehend the meaning of the sign. Implications of the findings for real-world roadworks zones are discussed.

Name company:	Rijkswaterstaat
External supervisor:	Ingeborg Haers
UU Supervisor:	Krista Overvliet

Visual field defects (VFDs) in neurologically impaired children, often resulting from stroke or tumors, can significantly hinder development and quality of life. Traditional diagnostic tools like Standard Automated Perimetry (SAP) present challenges in pediatric populations due to their reliance on sustained attention, clear communication, and motor responses. This internship project supports an ongoing study investigating pupil perimetry (PP), an objective, non-invasive method that measures pupillary responses to light stimuli using eye-tracking, as a potential alternative for assessing VFDs in children.

The project specifically explores a novel movie-watching pupil perimetry paradigm designed to enhance engagement in young participants. My role focused on collecting and analyzing data from healthy children, serving as controls. Three PP tasks were evaluated: gaze-contingent flicker PP (gcFPP), movie-watching (MW), and a combined condition (mw-gcFPP). Children's engagement was assessed using an adapted NASA-TLX scale presented in an interactive, child-friendly format. Visual field asymmetry and blind spot responses were also analyzed to evaluate the sensitivity and reliability of PP measures.

This internship contributes to understanding the feasibility of PP in pediatric research, with a particular focus on sensitivity to visual field differences and participant engagement. Findings may inform the development of more accessible diagnostic tools for young patients with neurological impairments.

Student Name: Matej Aluloski
Student ID: 6752071
Student Email Address: m.aluloski@students.uu.nl

Component: Internship
Title: Red Team Project: Analyzing & Interpreting a Massive Simulated Phishing Attack Dataset

Name of Employer: ING Bank
Name of External Supervisor: Stijn Meijer
Name of UUSupervisor: Marnix Naber

Abstract

This project analyzes a comprehensive phishing simulation dataset containing over 270,000 participant records across 52 behavioral and demographic variables, spanning multiple years of internal corporate security testing. Using data from 17+ distinct phishing templates deployed by an enterprise red team, we examine employee vulnerability patterns, training effectiveness, and victimization trends across several corporate locations and hierarchies.

This project examines training material efficacy by analyzing phishing susceptibility before and after provided training, as well as victimization rates across various job roles, pay grades, locations and departments. One key achievement includes the ability to quickly output descriptive statistical analyses of both individual employees and larger, categorized groups through several Python scripts, developed entirely by myself. Through their use, future red teams can quickly fetch granular data about both specific targets and broader groups, enabling better, more precise simulated attacks.

Name: Reinhard Heideman

Internship or thesis: Thesis

Title project: Bridging the gap: Do protocols influence gaze behaviour in remote bridge control?

Abstract:

This thesis explores whether a structured viewing protocol influences the gaze behaviour of remote bridge operators. Although such protocols are already being implemented in practice, empirical research on their effectiveness remains limited. To address this gap, a between-subjects eye-tracking experiment was conducted in a simulated remote bridge control setting, and these results were formally analysed. Participants either received structured viewing instructions or no such instructions. Three methods were used for this: transition probability matrices, gaze transition entropy and modified K-coefficient. Between the groups, there was a significant difference ($p < 0.000$) in the transition probability matrices. There was also a significant difference in gaze transition entropy between the groups, but no significant difference was found between the results of the modified K-coefficients. These findings provide insight into how viewing protocols can shape visual behaviour in safety-critical remote operations.

Name company or University: Utrecht University

UU Supervisor: Chris Janssen

Name Student: Fabienne de Mos

Internship or Thesis: Thesis

Title project: The Role of Advice Accuracy in Category Learning and Independent Performance

(Abstract, 100-150 words)

This study explored how advice accuracy affects learning in a categorization task. Sixty participants were randomly assigned to one of three conditions: no advice, low-accuracy advice, or high-accuracy advice. Participants categorized 254 cartoon-like humanoids (“Oomplets”) based on visual features. In the practice phase, advice was shown (except for the no-advice group), followed by a test phase with no advice for all groups to assess independent performance. Accuracy and reaction times were analysed using linear and generalized linear mixed-effects models. Results showed a significant interaction between Advice and Trial in the test phase. The high-accuracy group performed significantly faster and more accurately than the other groups. No differences were found between the low-accuracy and no-advice groups. These findings suggest that reliable advice improves learning and supports better independent performance, while inaccurate advice offers no benefit. The results highlight the importance of advice quality in transitions from guided to autonomous decision-making.

Name company or University: Utrecht University

(If applicable) **External supervisor:** NA (Auditor is Dr. Leendert van Maanen).

UU Supervisor: Dr. Baptist Liefoghe

Name Student: Maarten Verschoor

Internship or Thesis: Thesis

Title project: 'Proactive Response Inhibition and theta and ADHD'

My thesis was aimed to be partly a replication of a research that was already done by my supervisors and partly a new study. As a replication I investigated the link between proactive response inhibition and theta wave frequency and for the 'new study' I did research about the link between theta wave frequency and ADHD. For this I used EEG on about 50 real participants and was guided throughout the whole project that I did with a few other students of the ACP masters. It was really nice to put the learned skills of my bachelor into practice for once.

Name company or University: University of Utrecht

UU Supervisor: Sam van Bijn

Name Student: Annelore Jacobs

Internship/ thesis: Internship

Title project: AI as a Data Consultant

Abstract:

This internship report describes my role as a Business and Data Insight intern at Mindshare, During this internship I applied cognitive psychology theories to real-world media research.

The focus was on evaluating advertising effectiveness using both quantitative and qualitative methods. A key challenge emerged in analyzing open-ended survey responses, particularly detecting emotional nuances using AI tools like ChatGPT. The project showed limitations in AI's ability to correctly interpret sarcasm, implicit sentiment, and emotional ambiguity. This advisory report outlines how human cognition uses context, prior knowledge, and non-verbal cues—capabilities still lacking in AI systems. Based on this, I made four recommendations: combining AI with human oversight, training models on context-specific data, integrating psycholinguistic features, and flagging low-confidence classifications. These strategies aim to enhance the accuracy and efficiency of qualitative analysis in media research. The internship bridged theory and practice, highlighting the relevance of applied psychology in modern data consultancy.

Name company: Mindshare

External Supervisor: Irene Venings

UU supervisor: Sjoerd Stuit

Name Student: Manon Arns

Internship or Thesis: Thesis

Title Project: The relationship between self-reported stress and frontal theta power: an EEG study

Abstract:

Several previous studies provide evidence that stress has an effect on an individual's ability to inhibit their responses. Such an effect should be detectable in EEG measurements through a frontal theta power analysis, since frontal theta activity has been proven to be predictive for response inhibition performance. It is a marker of proactive control, which implies a state of 'readiness' of the brain to inhibit expected responses, before a stop-signal is presented. This makes frontal theta power a predictor of one's performance on response inhibition tasks. If one has high frontal theta activity, this results in more successful inhibition. This study aims to investigate the relationship between self-reported stress level and perceived frontal theta power. To study this association, two resting-state EEG measurements have been recorded and analyzed. Also, chronic stress was measured with a stress questionnaire. To draw conclusions, patterns were studied between questionnaire scores and perceived theta power.

(Results and conclusions to be added)

Name company or University: Utrecht University

External supervisor: -

UU supervisor: Leon Kenemans

By Ariaan Aberan Tabrizi
Supervisor: Jana Klaus
Utrecht University
Thesis project

Abstract

Sensory Processing Sensitivity (SPS) is a personality trait that is associated with enhanced emotional awareness and greater depth of information processing (Aron & Aron, 1997). However, there is still little research on SPS its relation to language prediction and linguistic memory. Hence why, in this study we want to investigate the following: **do individuals with high SPS have better prediction of emotional and neutral words compared to those with low SPS?**. In addition, this study will look at the following: **do individuals with high SPS have better word recall of emotional and neutral words than those with low SPS?** This study uses the Cloze task and a memory recall task to investigate this. In addition, the SPS score of each participant is measured with the Highly Sensitive Person Scale. In this study, participants showed significant slower reaction times for neutral context trials compared to negative and positive trials. However, there was no difference in reaction times for the memory task. There was also no evidence for SPS and emotion interaction. In conclusion, emotional valent context leads to faster language processing. Furthermore, SPS did not influence language prediction or memory in emotional contexts.

Name Student: Annemijn Koch

Internship or Thesis: Thesis

Title project: Sprinting Towards Sustainability: The Effect of Agile Ways of Working in Retail Sustainability Performance

As sustainability becomes increasingly central to business strategy, this thesis examines whether agile approaches to working, encompassing agile practices, leadership, and mindset, impact the relationship between sustainability strategy and performance in the retail sector. While agile emphasizes short-term adaptability, sustainability requires long-term commitment. Using a mixed-methods approach that incorporates literature reviews, surveys, interviews, and document analysis, the study evaluates three retail companies (2020–2024) through the Triple Bottom Line (TBL) framework, encompassing people, planet, and profit.

Findings confirm that strong sustainability strategies lead to better performance. Companies with higher agile maturity showed stronger alignment between their organizational practices and sustainability goals. While agile ways of working improved the emergence of a strong sustainability strategy, they did not affect the link between strategy and performance. This implies that, even if agility by itself does not ensure better sustainability results when combined structurally, it can support and improve sustainability strategies.

Name company or University: Capgemini Invent

(If applicable) External supervisor: Yasmin van den Anker & Sofie Ploegmakers

UU Supervisor: Sjoerd Stuit

Name Student:: Christopher Schrub

Internship or Thesis:

Title project: Performance Evaluation of Learning Processes Using Eye Tracking in Initial Pilot Training

Pilot training adheres to a highly standardized syllabus, which constrains opportunities for tailored instruction. One prerequisite for personalized training lies in the instructor's ability to monitor the trainee's decision-making whilst present in the cockpit. Eye tracking can offer new insights into student pilots' cognitive processes during flight tasks, potentially enabling more tailored training interventions. While previous research has established clear differences in gaze behavior between novices and experts, little is known about the transitional phase from low to higher levels of expertise. Understanding this transition is crucial to determine how eye tracking data can be effectively leveraged for training evaluation. We analyze eye tracking data from a longitudinal flight simulator study involving novices. We investigate the relationship between eye tracking metrics and performance. Preliminary results indicate that gaze behaviour begins to adapt after six weekly training sessions, suggesting early changes in visual strategies during the initial acquisition of flight skills.

Name company or University: Netherlands Aerospace Centre

(If applicable) External supervisor: Dirk Thijssen

UU Supervisor: Christoph Strauch

Name Student: Lisa D'hoore

Internship or Thesis: Internship

Title project: Social-Oriented vs Task-Oriented Conversational AI Agents: Differences In Likeability And Usage

(Abstract, 100-150 words)

On the CoVince platform, users have the opportunity to learn more about their soft and hard skills, vulnerability and resilience patterns by answering two questionnaires. To fully leverage the results of the questionnaires, a chatbot is at the disposal of the users in order to help clarify any questions the user might have, as well as suggest ways to, for instance, improve their less developed skills. The aim of the present research was to measure how the conversational style of the chatbot determined its likeability. More specifically, by conducting an A/B test, we investigated whether a social-oriented chatbot was more likeable than the present CoVince chatbot, as measured with the Godspeed Questionnaire. A t-test was conducted to determine any statistically significant differences between both groups.

Name company or University: CoVince

(If applicable) External supervisor: Richard van Tilborg

UU Supervisor: Sjoerd Stuit

Name Student: Chiara Battista (7373236)

Internship or Thesis: Internship (12,5 ECTS)

Title project: Strategies to foster engagement with a Mental Health Chatbot.

Despite the growing availability of Digital Mental Health Interventions (DMHIs), sustained user engagement remains a major challenge. This research explores how to enhance engagement with *Buddy*, an AI-powered mental health chatbot integrated into the Nexus platform by CoVince. A scoping literature review identified eight core engagement strategies, including personalisation, gamification, mood tracking, and media variety. Building on these findings, a mid-fidelity prototype was developed to test three selected strategies through a between-subjects A/B design. Twenty-four participants interacted with different chatbot versions and completed the User Engagement Scale – Short Form and open-ended feedback. While results are still being analysed, this study provides an evidence-based foundation for designing emotionally resonant and user-centred chatbot features, with the goal of supporting personal well-being through sustained digital interaction.

Name of company or University: CoVince

(If applicable) External supervisor: Richard Van Tilborg

(richard.van.tilborg@covince.com)

UU Supervisor: Marnix Naber (m.naber@uu.nl)

Name: Sana Yasmin

Internship or Thesis: Internship (12.5 ETCS)

Title project: Engaging Research Panel Members through Nudges: A Systematic Literature Review of Motivation and Reward Strategies

Company/University: Capgemini

External supervisor: Alankar Tayal (alankar.tayal@capgemini.com)

UU Supervisor: Sam Van Bijnen (s.vanbijnen@uu.nl)

Abstract

This project investigates how different types of rewards can motivate and encourage employees to participate in organizational initiatives, such as user interviews, usability testing, and surveys, activities essential for improving internal tools and systems. Despite the value of employee input, participation often remains low due to psychological and structural factors, including a lack of autonomy and misaligned incentives. Drawing on Self-Determination Theory (SDT), the research examines how intrinsic and extrinsic rewards, as well as their timing and design, influence motivation. In the context of Capgemini's effort to build a global UX research panel of 4,000 employees, this literature review aims to determine which reward strategies are most effective in encouraging employees to join and remain engaged in company-wide feedback initiatives.

EFFECTS OF INDOOR AIRCRAFT NOISE ON COGNITION



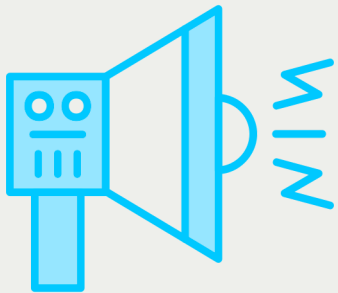
Universiteit
Utrecht



Accelerating
the future
of aerospace

Aleyandra Smitt

HOW DO DIFFERENT INDOOR AIRCRAFT NOISES AFFECT ANNOYANCE AND DISTURBANCE DURING DIFFERENT COGNITIVE STATES (RELAXED AND WITH A HIGHER COGNITIVE LOAD)?



PROBLEM:

The short-term consequences of indoor airplane noise on cognition are poorly understood.

WHAT DO WE WANT TO FIND OUT?

Hypothesis



A significant difference in annoyance, disturbance and coping is expected in indoor aircraft noise for participants, either in a relaxed condition or during a task requiring higher cognitive effort

- Sub-question 1:** What effect does noise intensity and type of aircraft have on annoyance and disturbance in a relaxed state?
- Sub-question 2:** What effect does noise intensity and type of aircraft have on a cognitively demanding task, such as a math task?
- Sub-question 3:** Is there a significant difference in coping with noise after a relaxed task or a math task?

RESULTS

- We found significant differences in annoyance and disturbance levels across the aircraft types, with specific aircraft types being more disturbing than others during certain tasks
- Noise tends to be more disturbing during a math task than a colouring tasks
- No difference found in coping during the two tasks



Figure 2. A 360 degree picture of the LOUNGE facility at NLR, Amsterdam

WHY IS THIS INTERESTING TO US?



- The results could aid in urban planning, policy-making and mitigating noise.

TYPES OF NOISE UTILIZED

Type of aircraft	Sound level	Duration of flyover
Airbus A330	80.6 dBs	50 seconds
AW139	74.07 dBs	50 second
EC 135	68.05 dBs	50 seconds
B737 MAX	64.06 dBs	50 seconds
DRONE M600	63.72 dBs	50 seconds



Figure 1. Type of aircrafts and their respectable peak sound levels presented in the study

CONDitiONS

- **"Relaxed" state**
The participants will be colouring while listening to aircraft sounds
- + **"Higher cognitive task"**
The participants will complete math tasks while listening to aircraft sounds



MEASURMENTS

- Surveys
- Heart rate (using a Polar band)
- Sweat response (using electrodermal activity)



PARTiCiPANTS

- 33 participants tested
- 18 of them lived close to the airport
- 14 of them worked in aviation

Name Student: Sean Garvey

Thesis

Project Title: Influence of Low-Level Image Properties on Access to Awareness between
Clothed and Nude Bodies

The term 'sex sells' is a reflection of how often nude bodies are used to grab attention in advertising, but what makes them so effective? While the general consensus indicates a biological effect: humans being hard-wired to detect potential mates and competitors, there may be more to the story. Differences in low-level image features could be playing a key role in why nude bodies reach visual awareness faster than clothed bodies. This forms the basis of our research question: Are the differences in time required to reach visual awareness between nude and clothed bodies explained by low-level image features? To answer this question we used a behavioural simultaneous masking task to collect data, and a model developed based on protosc toolbox to analyse the effect of image features. In this study, it is hypothesised that access to awareness will be unlocked faster by images of nude bodies than images of clothed bodies. It is also predicted that this difference will be influenced by differences in low-level image properties.

Utrecht University

Supervisor: Sjoerd Stuit

Name Student: Stefan Popma

Internship or Thesis: Thesis

Title project: The Psychology of Meat Misconceptions: Investigating Cognitive Biases Behind Knowledge levels on Meat Related Misconceptions

This study aims to investigate cognitive biases behind knowledge levels on meat related misconception. Despite the well-documented negative consequences of meat consumption on animal welfare, the environment, and public health, people still continue to eat meat. This might be explained by the many misconceptions on meat, which may be attributed to psychological mechanisms like social vigilantism, overconfidence and cognitive dissonance. These factors can lead to polarized conversations between meat-eaters and non-meat-eaters, in which differing perspectives are perceived as competing truths.

The study specifically aims to investigate whether limited knowledge contributes to the acceptance of factory farming and meat consumption, or in contrast, whether such knowledge gaps conveniently align with psychological mechanisms. A knowledge test and accompanying questionnaire will be used to measure beliefs, knowledge levels, and attitudes. By uncovering the psychological and demographic factors behind misconceptions on meat, this research seeks to improve effective communication around plant-based food.

Name company or University: Utrecht University / Radboud University

External supervisor: Roos Vonk

UU Supervisor: Samson Chota

Name Student: Junyan Li

Internship or Thesis: Thesis

Title project: The Head-in-world and Eye-in-head Orientation in Triadic Conversation

Abstract: Gaze behavior is essential in daily activities and complex social interactions. Beyond eye movements, head direction significantly influences gaze, especially in real-world settings requiring large spatial movement. While head direction often indicates gaze, people may decouple eye and head movements in socially sensitive situations to conceal attention. Leadership roles in group interactions can also shape gaze behavior. This study used wearable eye trackers and ArUco markers to examine "head-in-world" and "eye-in-head" movements during three-party conversations. 33 participants in 11 groups completed a 15-minute collaborative survival task while their eye and head movements were recorded. Results revealed attention allocation patterns and how head and eye cues jointly indicate focus. Group gender composition influenced gaze behavior, and leaders showed distinct gaze patterns from followers. Future research should further explore demographic effects on nonverbal communication by improving stimulus distribution, head motion capture accuracy and sample diversity.

Name company or University: Utrecht University

(If applicable) External supervisor: /

UU Supervisor: Roy Hessels

Name student: Veerle Hoogerwerf

Thesis: Numerosity

Utrecht University

UU Supervisor: Prof. Dr. Maarten van der Smagt

The human visual system can rapidly estimate the number of items in a scene—a capacity known as numerosity perception. Although the phenomenon of numerosity has been studied for decades, the location where numerosity processing takes place remains unclear. In particular, it is unknown whether numerosity is processed at early monocular stages (e.g., in subcortical structures or early V1 layers) or at later binocular stages of the visual stream. To address this, we used an interocular transfer (IOT) paradigm to test whether adaptation to numerosity occurs before or after binocular integration. Seventeen participants adapted to dot arrays with either low (20) or high (80) numerosity in one eye, then judged the numerosity of test stimuli shown to either the same (monocular viewing) or the opposite eye (IOT). We observed a robust adaptation aftereffect in the IOT condition. In contrast, no significant effect was found in monocular viewing conditions. There was no interaction between viewing condition and numerosity adaptation, which was mainly caused by the monocular viewing condition. Furthermore there was no interaction between monocular and interocular viewing conditions. These results suggest that numerosity is encoded later in the visual processing stream. This supports the view of numerosity as a high-level perceptual property, and contributes to ongoing debates about its neural origins.

Name Student: Novela Perić

Internship or Thesis: Thesis

Title project: The Future of Virtual Humans: a Delphi Study on the Emerging Trends and Implications of Human-Virtual Human Interactions

The advancements in modern technology are shifting our daily lives, one such example are the developments in the field of virtual environments, specially virtual humans. Virtual humans are seen as highly realistic digital representations that simulate human appearance and behaviours (Burden & Savin-Baden, 2019). Despite their growing presence and use, little is known on how their implementation will influence our everyday lives, and with that how to properly govern them. This study aims to find insights on how the advancements in VHs will influence human life in the future and what can be done now in order to shape an ethically sustainable, symbiotic human-virtual human future. A Delphi study, gathering expert opinions has been adapted as the most adequate methodology to explore future scenarios. Results remain unknown, as the data collection phase is still open, however a panel of 30 experts across Europe have already shared their opinions on 30 statements regarding these topics.

Name company or University: Cradle, Breda University of Applied Sciences

(If applicable) External supervisor: Igor Mayer

UU Supervisor: Samson Chota

**Aircrafts and perceived annoyance: evaluating the psychological effect of
simulated aircraft sounds in real world vs VR settings**

Andreea Maria Birgaoanu

SNR: 9796819

Master of Applied Cognitive Psychology

Utrecht University

Royal Netherlands Aerospace Centre

Internal supervisor: Janssen

External supervisor: Roalt Aalmoes

Abstract

Aircraft noise disrupts sleep and cognition and threatens health (Benz et al., 2022; Miedema & Oudshoorn, 2001). Different aircraft types may cause varying levels of annoyance, influenced by factors like loudness, pitch, and attitudes towards aircraft (Sieben et al., 2024; Yu et al., 1997). Also, perception is highly shaped by our surroundings (Locher et al., 2018). This study examined the link between aircraft noise and annoyance across indoor and outdoor environments inside two contexts: lab LOUNGE and Virtual Reality (VR), using a within-subjects design ($N = 33$). Participants rated annoyance across conditions. Recruitment targeted adults via social media, university ads, and NLR. Analyses involved various ANOVAs and a correlation regarding annoyance ratings and noise sensitivity. Results showed drones were slightly more annoying than airplanes in the lounge; no differences emerged in VR. LOUNGE showed more annoyance in outdoor settings. Follow-up research should explore aircraft annoyance in various environments.

Name Student: Celine Rosen

Internship or Thesis: Thesis

Title Project: The Role of Brand Memory Strength in Consumer Decision-Making for Food Products

Consumer decision-making is shaped by brand memory, with familiar brands more likely to be chosen. A-brands (national brands) tend to be more easily recognised than B-brands (private labels). This study examined whether prior exposure to unpackaged products affects later recognition when packaging is introduced, focusing on brand memory strength while controlling for brand familiarity and consumption frequency. In an online experiment, Dutch participants (N = 30) first viewed unpackaged food products and later completed a recognition task involving the same products with packaging (A- and B-brands). No significant differences were found between A-brands and B-brands in recognition accuracy, reaction time, familiarity, or frequency. A mixed-effects logistic regression revealed that shorter reaction times and higher brand familiarity significantly predicted higher recognition accuracy, while consumption frequency had no significant effect. The small sample size limits generalisability, but nonetheless, the findings highlight the importance of prior brand knowledge in shaping consumer decision-making.

Name Company or University: Utrecht University

(If applicable) External Supervisor: /

UU Supervisor: Martijn Mulder

Name Student: Toos van Dijken

Internship or Thesis: Thesis

Title project: Touch and the Post-Auricular Reflex: Comparing Self, Social, and Observed Touch

Abstract

Background Touch is a fundamental component of human interaction, which can play a crucial role in emotional regulation, social bonding, and well-being. The post-auricular reflex (PAR) has proven to serve as an implicit measurement of positive affect, and therefore might also be a physiological marker of the pleasantness of touch. The aim of this project is to investigate whether social touch elicits a stronger PAR and higher subjective pleasantness ratings compared to self-touch.

Methods A sample of participants ($n=75$), take part in a $2 \times 2 \times 2$ within subjects experimental design. The conditions are self-touch versus social touch, touch by hand or touch by make-up brush, and touch on the participants real arm or on a fake arm. Subjective pleasantness ratings are given after each touch experience and the PAR is measured during each condition.

Results A linear mixed-effects model is conducted and shows significant differences in PAR across touch conditions ($<.001$).

Conclusions It is predicted that the social touch conditions will elicit a stronger PAR and align with higher subjective pleasantness ratings of the participants.

K E Y W O R D S Pleasantness of Touch, Social touch, Self-touch, Post-Auricular Reflex

Name company or University: Utrecht University

UU Supervisor: David Terburg